Note 5.4 Inner Product Spaces

Definition and Examples

Definition

Inner product is an operation on a vector space V that corresponds every pair of vectors $\mathbf{x}$ and $\mathbf{y}$ in V, denoted by $\langle \mathbf{x}, \mathbf{y} \rangle$, satisfying that

- The inner product of two identical vector are nonnegative.
- commutative law
- linear combination.
  and the vector space with an inner product is inner product space.

Inner Product in Different Vector Spaces

For $\mathbb{R}^n$, $\mathbb{R}^{m\times n}$, $C[a,b]$, P_n.

Basic Properties of Inner Product Spaces

Length (Norm)

$$||\mathbf{v}|| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$$

Orthogonal

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0$$

Theorem 5.4.1 The Pythagorean Law

For two orthogonal vectors $\mathbf{u}, \mathbf{v}$ in an inner product space V,

$$||\mathbf{u} + \mathbf{v}||^2 = ||\mathbf{u}||^2 + ||\mathbf{v}||^2$$

Frobenius Norm

$$||A||_F = (\langle A, A \rangle)^{\frac{1}{2}} = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2 \right)$$

Definition of Scalar Projection and Vector Projection

The scalar projection of $\mathbf{u}$ onto $\mathbf{v}$:

$$\alpha = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{||\mathbf{v}||}$$

The vector projection of $\mathbf{u}$ onto $\mathbf{v}$:

$$\mathbf{p} = \alpha \left(\frac{\mathbf{v}}{||\mathbf{v}||} \right) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}$$

Remarks

- $\mathbf{u} - \mathbf{p}$ and $\mathbf{p}$ are orthogonal
- $\mathbf{u} = \mathbf{p} \Leftrightarrow \mathbf{u}$ is a scalar multiple of $\mathbf{v}$.

Theorem 5.4.2 The Cauchy-Schwarz Inequality

$$||\langle \mathbf{u}, \mathbf{v} \rangle|| \leq ||\mathbf{u}|| \, ||\mathbf{v}||$$

and equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

It can be proved by the projection and Pythagorean law.

And we can make a unique angle θ such that

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{||\mathbf{u}|| \, ||\mathbf{v}||}$$

Norms

Definition of Normed Linear Space

A normed linear space V for each vector $\mathbf{v} \in V$, there is an associated real number $||\mathbf{v}||$, the norm, satisfying

- nonnegative
- scalar multiple($||\alpha \mathbf{v}|| = |\alpha| \, ||\mathbf{v}||$)
- triangle inequality

Definition of Distance

The distance between $\mathbf{x}$ and $\mathbf{y}$ is defined to be the number $||\mathbf{y} - \mathbf{x}||$.